

What are Self-Aware and Self-Defending Adaptive Networks?

A self-defending adaptive network is the next generation network that defends itself from security breaches. It is a system that knows and recognises the level of threat faced by an intrusion across a network. It uses a new method of machine learning to track threats, which can include ransomware, code hijackers, intrusion and illegal entry, theft, and unauthorised use. Such a network uses a set of modules that can be added or removed to operate and achieve specified goals. It can change its behaviour by modifying network devices.



Adaptive networks with AI are used for automated service provisioning. Here, network providers automate manual service life cycle processes. These processes are automated using packet and optical networks. Packet and optical networks are built using a software defined networking based automated platform, which is multi-layer and multi-vendor based and adopts DevOps processes. For

instance, a network provider can automate the delivery of its wavelength services and plans, and can automate to extend this platform to other services using a phased approach.

Proactive network assurance is another area where AI based adaptive networks can be used. The network providers need to identify and correct as many network issues as they can foresee and predict. This helps in increasing network reliability and

delivery with specified SLAs. AI based adaptive platforms improve the customer experience as they have features related to pre-emptive network maintenance across the optical, Ethernet and IP wide area networks (WANs). AI based automated platforms have network health prediction capabilities. Along the same lines, machine learning based analytics can predict the likelihood of a network node's failure in a given timeframe.